

The Pragmatic Skripsi: How I Wrote My Undergraduate Thesis Using FOSS on openSUSE

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I am open for new opportunities, mainly Backend, Devops, SRE.

Before we start...

I want to ask y'all first

What You'll Learn

- Managing expectations and avoiding productivity traps while writing your thesis
- Assessing your unique software requirements to select the right tools for your workload
- Essential openSUSE installation and verification techniques so you can manage your setup self-sufficiently

Internal & External Problem



The Tinkering Trap (Procrastination by Tooling)

- Learning Curve Friction
- Compatibility Anxiety
- The Tinkering Trap: Procrastination by tooling (no, tweaking dotfiles and browsing r/unixporn doesn't count as writing)
- FOSS Dogmatism / Purism

External

Mostly from your supervisor and your college requirements.

External

- Don't force them to follow you.
- Use a universal file format like PDF.
- Demonstrate to your supervisor that open-source tools deliver identical results.

Plan and Map Feature You will Need



What will you need?

You need to be specific to what feature you will need. For example you need some kind of pagination in Word. You need specific Macros that only work for Excel. It's up to you.

Just try it first in your ‘usual’ apps

Research for Alternative

Once you outline your needed features, search for open-source alternatives that match them.

Use AI (responsibly)

Sometimes you need to research, how to do this, how to do that in FOSS App.

AI is extremely helpful. You don't need to ask forum and hope someone answer you like in the old day.

Be Pragmatic

You don't have to use FOSS for everything.

Try FOSS first and see if it can cover what you need.

If it works, great. You can avoid paying for expensive software.

If it doesn't, just use the proprietary app.

The goal is to get your thesis done, not to use FOSS at all costs.

Case Study: My *Skripsi* Workflow

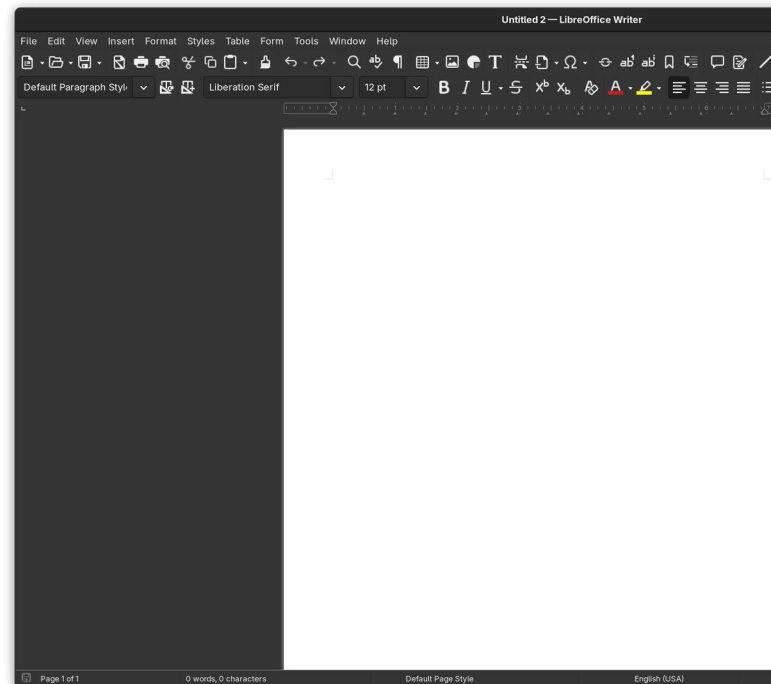


The Editor

First I thought that LaTeX would be good for me.

But then I realized I can't easily show the editor when consulting with my supervisor.

So I chose LibreOffice Writer.

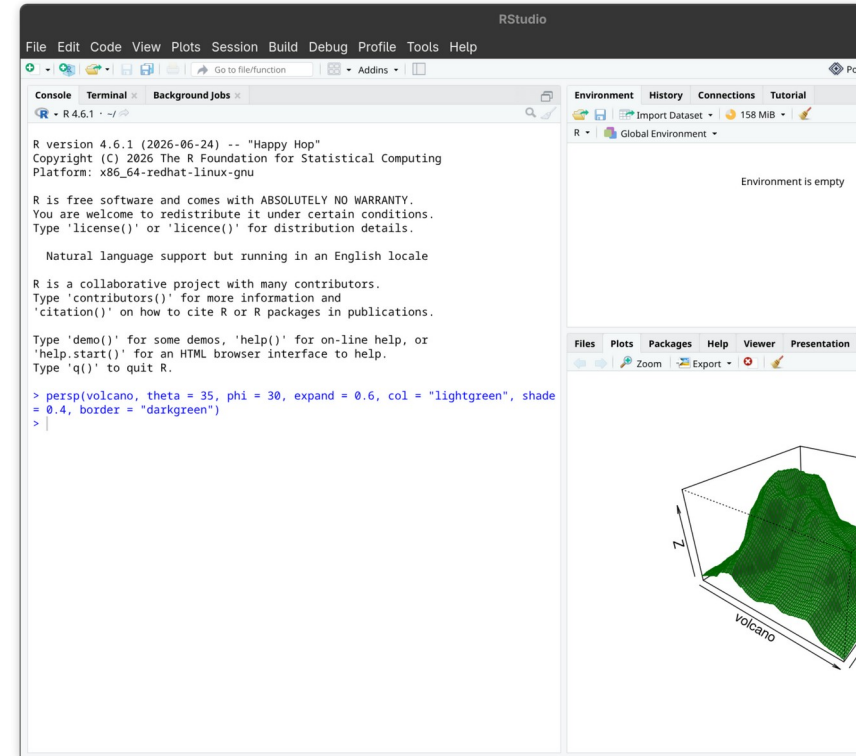


The Powerhouse

I needed PLS-SEM analysis.

The safe way is SmartPLS.

But I use Rstudio.



Classroom Companion: Business

Joseph F. Hair Jr. · G. Tomas M. Hult ·
Christian M. Ringle · Marko Sarstedt ·
Nicholas P. Danks · Soumya Ray

Partial Least Squares Structural Equation Modeling (PLS-SEM) Using R

A Workbook

OPEN ACCESS

 Springer

This book is free too, btw.

So I use Rstudio

<https://github.com/sem-in-r/seminr>

The Bibliography

Just use Zotero.

- Flawless LibreOffice Writer Integration
- CSL Engine & Custom Styles: Handles strict local campus citation guidelines and switches between styles (APA, IEEE, Harvard) in one click.
- One-Click Browser Capture: Instantly extracts full metadata and PDFs from Google Scholar, JSTOR, or journal sites.

The Hands-on



How to Install Something

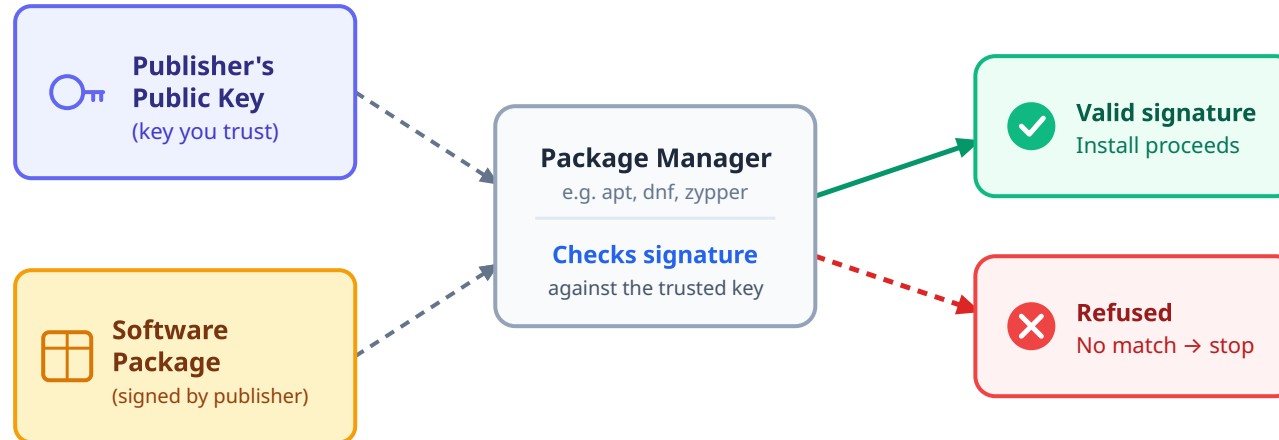
- zypper
- flatpak
- Manual binary Installation
- Compile it yourself

Why Zypper

Flatpaks currently require complex integration steps for LibreOffice plugins due to containerization.

It possibly could. But too much work.

Added once by you or your distro



Verifying (the good practice)

<https://posit.co/code-signing>

```
# import your key  
nano posit-new.key
```

```
# see the inside  
gpg --show-keys --with-subkey-fingerprints posit-  
new.key
```

Verifying (the good practice)

```
sudo rpm --import posit-new.key
```

```
rpm -Kv rstudio-2026.09.0-174-x86_64.rpm
```

Installing the Downloaded App

```
sudo zypper in ./rstudio-2026.09.0-174-  
x86_64.rpm
```

Installing the Downloaded App

```
mkdir -p ~/.local/opt ~/.local/bin  
~/.local/share/applications
```

```
mv Zotero_linux-x86_64 ~/.local/opt/zotero
```

```
~/.local/opt/zotero/set_launcher_icon
```

Installing the Downloaded App

```
ln -s ~/.local/opt/zotero/zotero.desktop  
~/.local/share/applications/zotero.desktop
```

```
ln -s ~/.local/opt/zotero/zotero  
~/.local/bin/zotero
```

Installing the Downloaded App

Before you open the Zotero!

```
sudo zypper in java-21-openjdk
```

Thank You



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